COMP 371 • Object Oriented Modeling and Design

Project Checkpoint 1

Scope and Project Proposal

Due	Submission	Purpose
Thu. Oct. 22, 2026 · 11:59 p.m. PT	Export completed template to PDF	Formative project review

This checkpoint becomes the starting material for your later project documentation. Revise and carry it forward rather than treating it as a separate report.

1. Project Identification

Field	Response
Project title	Study Room Reservation System
Work mode	☑ Individual ☐ Team (2–5)
Student / team members	Julia Wosnack 300114746
Git repository (optional at CP1)	https://github.com/lilypad-ios/Comp371-project https://github.com/users/lilypad-ios/projects/1/views/1
Date	November 11, 2026

2. Scope Expectations

If you are…	Plan for…
Individual	4–5 meaningful features; at least 2 should involve a business rule, decision, or state change.
Team (2–5)	5–7 meaningful features; at least 3 should involve a business rule, decision, or state change.

3. Problem Statement

Students currently have to call (or walk-in) to each location to ask for the next month's schedule, and to make study room bookings, changes and cancellations. They sometimes have to cancel last-minute, and end up having to pay no-show fees because the system is not correctly updated. Students are not informed about changes or cancellations mid-week. Changes to the system are not logged, so a mistake (such as removing the wrong student from a room booking) cannot be undone by the booking agents. Staff bookings override student bookings, and often these overriding bookings are not done correctly, and the student is not informed of the change. The university requires a system that assists (and informs) students, staff, and booking agents with their bookings, booking changes, and cancellations. The system should show study room availability (by room, timeslot, or building), up to a month in advance, and it needs to keep a log of booking, changes, and cancellations to assist booking agents.

4. Intended Users / Candidate Actors

User / candidate actor	What they need to do / Goals
Student actor	When interacting with the system, a student expects to be able to: • view and book available study room timeslots up to a month in advance • search for these study rooms by room, timeslot, or building • view, make changes to, and cancel their currently booked study room timeslots • make a single booking for an available timeslot • cancel timeslots up to the last minute before the timeslot booking begins • be notified of any changes or cancellations made to their booked timeslots
Staff actor	When interacting with the system, a staff member expects to be able to: • view and book all available and unavailable (due to a student booking) study room timeslots up to a month in advance (staff are able to see extended availability) • view and book over student bookings when necessary • search for these study rooms by room, timeslot, or building • view, make changes to, and cancel their currently booked study room timeslots • make multiple bookings for different study rooms during the same timeslot • cancel timeslots up to the last minute before the timeslot booking begins • be notified of any changes or cancellations made to their booked timeslots
Booking agent actor	When interacting with the system, the booking agent expects to be able to • search through all available and unavailable timeslots by room, timeslot, building, student id, or staff id booking • book, make changes to, and cancel bookings on behalf of students and staff • view which student or staff member is booked for a given timeslot • view the logs of all bookings, changes, and cancellations made • edit existing bookings to adjust for mistakes • be notified of student bookings that are overridden by staff bookings, so they can search for an alternative for the student

5. Proposed Features

A meaningful feature represents a user goal/workflow and normally involves a rule, decision, state change, object interaction, or persistence effect. Supporting CRUD/query operations may be listed but should not dominate the project.

#	Proposed feature	Core workflow or supporting operation?	Brief description
F1	The system shall allow users to search for available study rooms by room, timeslot, or building, up to a month in advance. The system shall allow staff to additionally view student bookings, and booking agents to additionally view all timeslots and search by student/staff id.	Core workflow	All users: search by available room, timeslot, or building, to see available timeslots for study rooms up to a month in advance. Staff: see timeslots booked by students. Booking agents: see all timeslots.
F2	The system shall allow users to book available timeslots for study rooms, and booking agents to book on behalf of staff/students. The system shall allow staff and booking agents to override student bookings, when booking an overlapping timeslot for the same study room.	Core workflow	All users: book available timeslots. Staff and booking agent: override student bookings.
F3	The system shall identify students, staff, booking agents, study rooms, bookings, and booking timeslots by a unique identification number.	Supporting operation	All users, study rooms, bookings, and booking time slots have unique identification.
F4	The system shall allow students and staff to cancel and make changes to their booking, before the timeslot begins. The system shall also allow booking agents to make changes or cancellations for student and staff bookings.	Core workflow	Students/staff: make changes to and cancel their bookings. Booking agent: make changes to and cancel bookings on behalf of students/staff.
F5	The system shall create a log of all new bookings, changes to bookings, booking cancellations, and overridden bookings. For overridden bookings, the system shall make a log entry of the new staff booking and the overridden student booking.	Supporting operation	The system will create logs for booking information.
F6	The system will keep booking log entries for booking agents to view which include the student/staff/booking agent identification number, date/time, study room id, booking id, and booking timeslot id.	Supporting operation	The system will keep logs for booking agents to view.
F7	The system must only allow 1 booking per study room per timeslot, and study room timeslot bookings must not overlap. It must not allow students to have more than 1 booking at the same time, but staff may have bookings for multiple different rooms at the same time.	Supporting operation	All users: 1 booking per study room per timeslot, and study room timeslot bookings must not overlap. Students: 1 booking at one time. Staff: multiple bookings at one time.

6. Initial Business Rules

Identify 2–4 initial rules. These are provisional and may change after use-case/design work.

Rule	Description
BR-1	The system must use the university's existing identity service for authentication.
BR-2	A "booking" means that there is only one associated student/staff identifier per study room per time slot.
BR-3	A user of the system includes students, staff, and booking agents, different users are assigned different booking privileges within the booking system.
BR-4	The log with the booking information is created by the system will be interpreted by an outside system, which emails the required recipient. The system will not send out emails.

7. Scope Boundaries

In scope	Out of scope
Allowing system access to authenticated users	User authentication and ensuring active status
Overriding student bookings	Finding a new booking for the student
Logging cancelled bookings	Cancellation Fees and tracking no-shows to bookings
Logging new bookings, changes to bookings, cancelled bookings, and overridden bookings	Sending out emails
A single booking per room	Multiple bookings per room
System backend functionality	Front end user interface

8. Initial Risks / Uncertainties

Identify one or more aspects you are currently unsure about (scope, data, workflow, technology, team coordination, etc.).

I'm a bit worried about database setup, just wondering if I'm going about this correctly. For the database planned setup, I think I need 6 tables: **Users** (user_id, user_role, user_name), **Buildings** (building_id, building_name), **Study Rooms** (room_id, building_id), Timeslots (timeslot_id, room_id, start_date_time, end_date_time, status), **Bookings** (booking_id, timeslot_id, user_id, room_id, booking_status), **Logs** (log_entry_id, booking_id, action, action_user_id). I am unsure whether or not to include the user interface, I was going to, but then I realized that it makes the project seem a lot larger, even though it would be simple to implement.

9. Questions for the Instructor

1. Does the database planned setup look approximately right?

2. If I don't include a user interface, what are you expecting from the program, an example list of entries?

3. _____

Before Submitting

☑ Scope matches the selected work mode.

☑ Feature list contains real workflows, not only CRUD.

☑ Business rules are visible.

☑ Out-of-scope items prevent uncontrolled expansion.

☑ Questions identify the feedback you most need.

Instructor Review

☐ On track

☐ Revise before next stage

☐ Instructor discussion recommended

Feedback / priority actions:

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